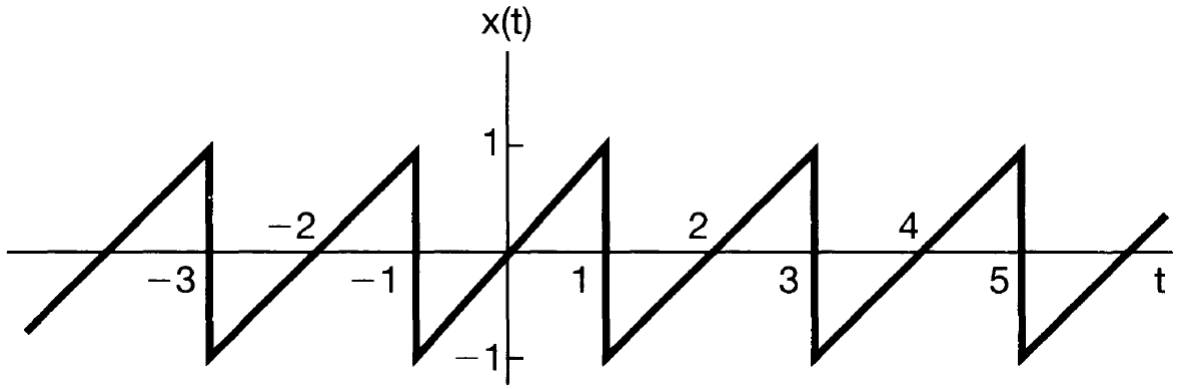
Continuous Time Fourier Series and Properties

Semester 2, 2025/2026



Exercise

Determine the Fourier series representations for the following signals $x(t)$.



Steps to solve

1. Determine x(t) for 1 period and choose segment for 1 period.
2. Find a_0 for $k = 0$
3. Find a_k for $k \neq 0$

Answer:

$$a_0 = 0$$

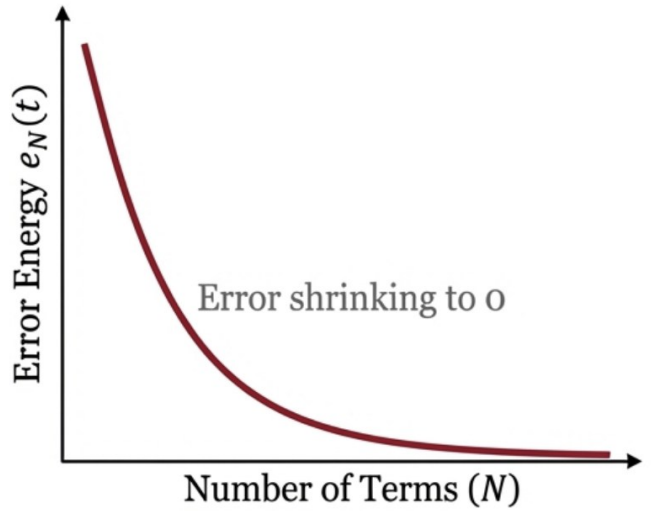
$$a_k = \frac{j(-1)^k}{k\pi}$$

Convergence of Fourier Series

- Fourier series can be used to represent an extremely large class of periodic signals **but not all**.
- One class of periodic signals that are representable through the Fourier series is those signals which have **finite energy over a single period**, i.e., signals for which

$$\int_T |x(t)|^2 dt < \infty$$

$\Rightarrow$ The energy E_N in the approximation error, converges to 0 as we add more and more terms ($N \to \infty$)



Dirichlet Conditions for Convergence

Guarantees that $x(t)$ equals its Fourier series representation, except at isolated values of t for which $x(t)$ is discontinuous.

Condition 1. Over any period, $x(t)$ must be **absolutely integrable***;* that is,

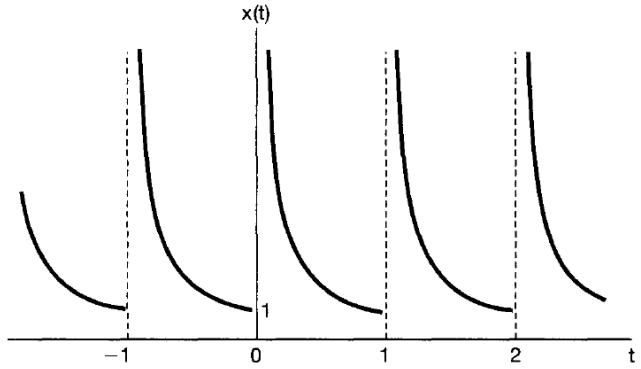
$$\int_T |x(t)|dt < \infty$$

Condition 2. In any finite interval of time, $x(t)$ is of bounded variation (**finite number of maxima and minima** within any finite interval)

Condition 3. In any finite interval of time, there are only a **finite number of finite discontinuities**.

Example: Signals that Violate Dirichlet Conditions

1. Infinite Discontinuity

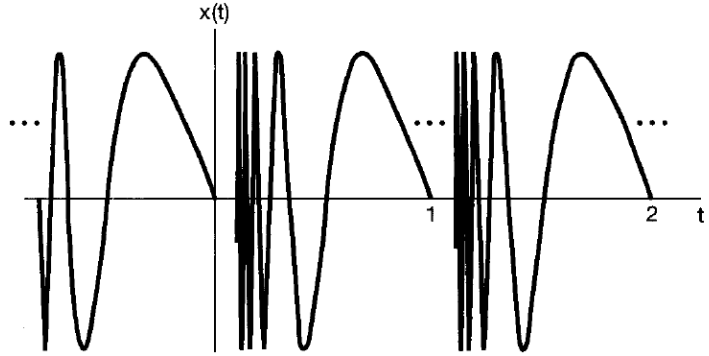


- Each period contains a vertical asymptote where the signal approaches $\pm\infty$.
- The value of $x(t)$ becomes unbounded near those points.

→ The signal is not absolutely integrable

→ The discontinuity is infinite

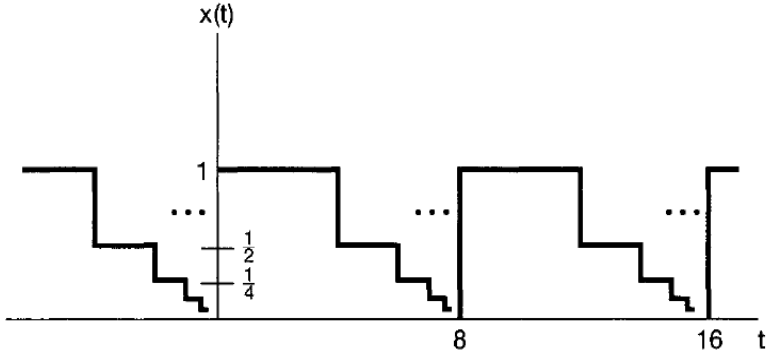
2. Infinite Oscillations



- Near certain points the signal oscillates infinitely fast.
- The waveform contains infinitely many peaks and valleys in a finite time interval.

→ There are infinitely many extrema.

3. Infinite Discontinuity

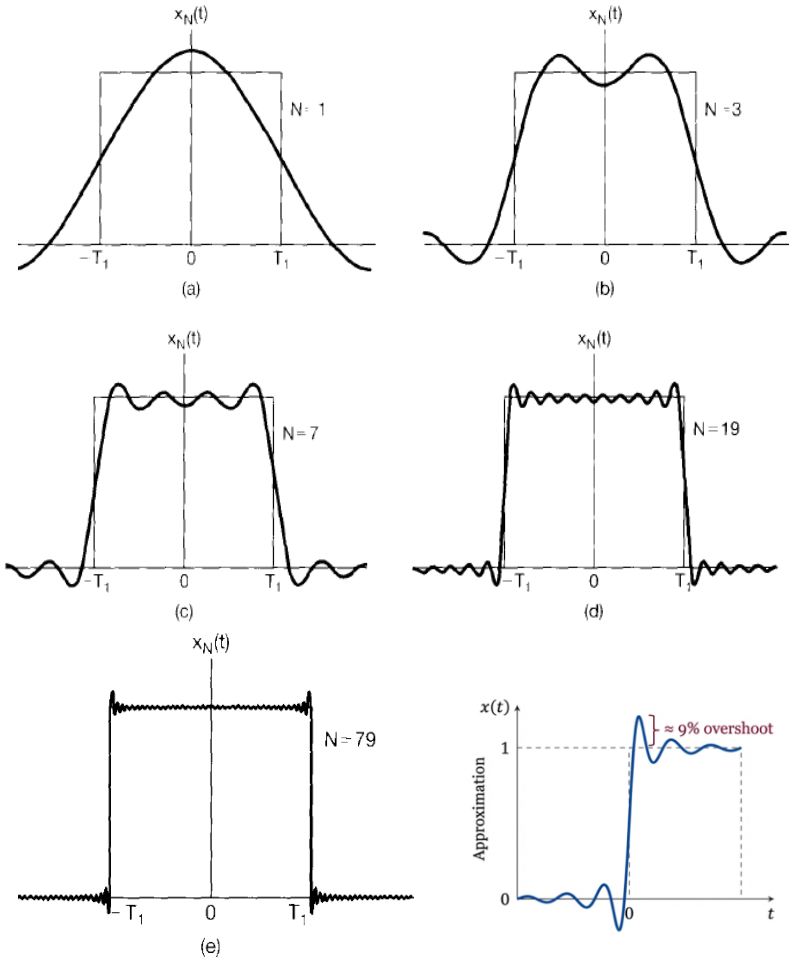


- The waveform is a staircase whose steps keep subdividing.
- The signal contains infinitely many jump discontinuities within one period.

→ The discontinuity is infinite

Gibbs Phenomenon

- For a periodic signal with a finite number of discontinuities in each period, the Fourier series representation equals the signal everywhere except at the **isolated points of discontinuity**, at which the **series converges to the average value of the signal on either side of the discontinuity**.
- As N increases, the ripples in the partial sums become compressed toward the discontinuity, but for any finite value of N, the **peak amplitude of the ripples remains constant**.



Convergence of the Fourier series representation of a square wave

Properties of Continuous-Time Fourier Series

Property	Periodic Signal	Fourier Series Coefficients
	$x(t)$, $y(t)$ Periodic with period T and fundamental frequency $\omega_0 = 2\pi/T$	a_k, b_k
Linearity	$Ax(t) + By(t)$	$Aa_k + Bb_k$
Time Shifting	$x(t - t_0)$	$a_k e^{-jk\omega_0 t_0} = a_k e^{-jk(2\pi/T)t_0}$
Frequency Shifting	$e^{jM\omega_0 t} = e^{jM(2\pi/T)t} x(t)$	a_{k-M}
Conjugation	$x^*(t)$	a^*_{-k}
Time Reversal	$x(-t)$	a_{-k}
Time Scaling	$x(\alpha t), \alpha > 0$ (periodic with period T/α)	a_k
Periodic Convolution	$\int_T x(\tau)y(t - \tau)d\tau$	$Ta_k b_k$
Multiplication	$x(t)y(t)$	$\sum_{l=-\infty}^{+\infty} a_l b_{k-l}$
Differentiation	$\frac{dx(t)}{dt}$	$jk\omega_0 a_k = jk\frac{2\pi}{T}a_k$
Integration	$\int_{-\infty}^{t} x(t)\,dt$ (finite valued and periodic only if $a_0 = 0$)	$\left(\frac{1}{jk\omega_0}\right)a_k = \left(\frac{1}{jk(2\pi/T)}\right)a_k$

Property 1: Linearity

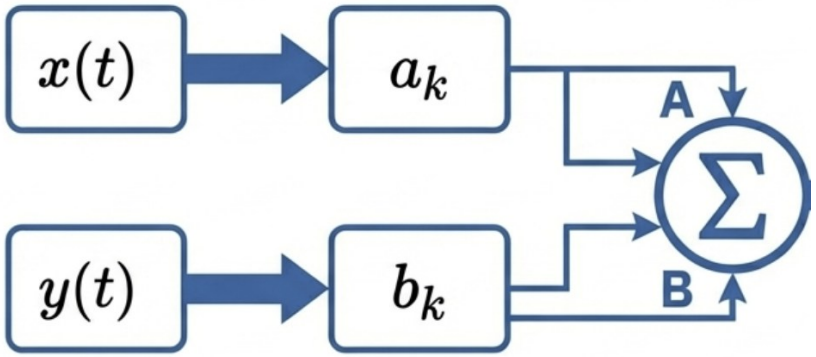
Let $x(t)$ and $y(t)$ denote two periodic signals with period T and which have Fourier series coefficients denoted by a_k, and b_k respectively.

$$x(t) \overset{\mathcal{FS}}{\leftrightarrow} a_k$$

$$y(t) \overset{\mathcal{FS}}{\leftrightarrow} b_k$$

Then

$$z(t) = Ax(t) + By(t) \overset{\mathcal{FS}}{\leftrightarrow} c_k = Aa_k + Bb_k$$



Property 2: Time Shifting

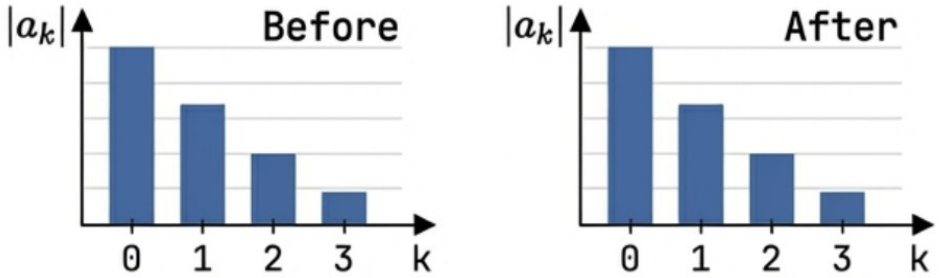
If

$$x(t) \overset{\mathcal{FS}}{\leftrightarrow} a_k$$

Then

$$x(t - t_0) \overset{\mathcal{FS}}{\leftrightarrow} e^{-jk\omega_0 t_0} a_k = e^{-jk(2\pi/T)t_0} a_k$$

When a periodic signal is shifted in time, the magnitudes of its Fourier series coefficients remain unaltered ($|b_k| = |a_k|$).



Property 3: Time Reversal

If

$$x(t) \overset{\mathcal{FS}}{\leftrightarrow} a_k$$

Then

$$x(-t) \overset{\mathcal{FS}}{\leftrightarrow} a_{-k}$$

- If $x(t)$ is even, its Fourier series coefficients are also even ($a_{-k} = a_k$)
- If $x(t)$ is odd, its Fourier series coefficients are also even ($a_{-k} = -a_k$)

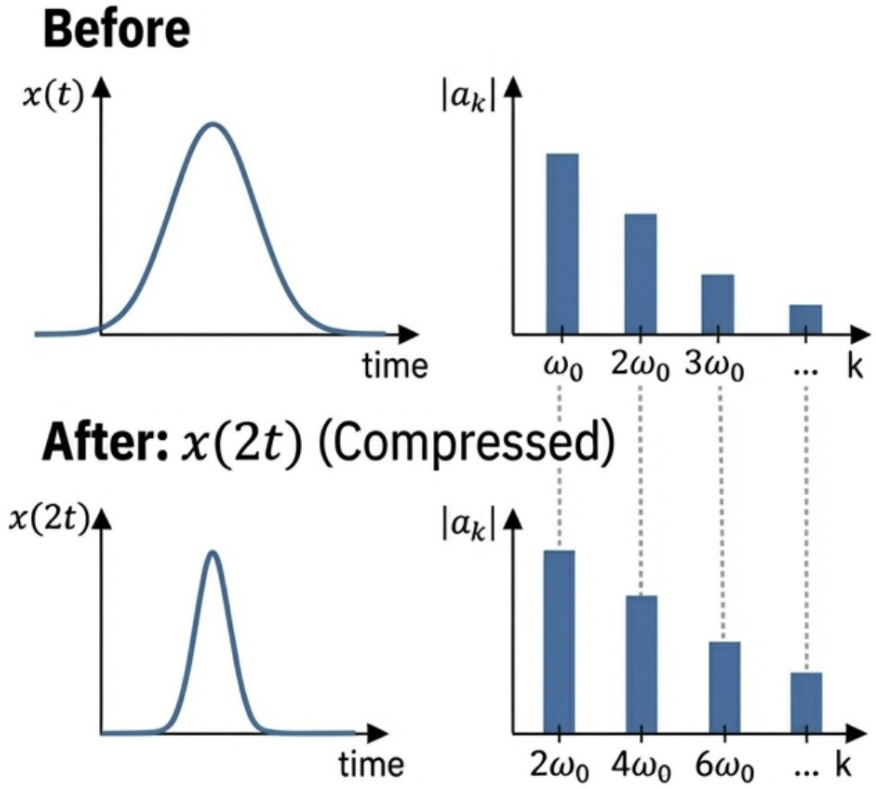
Property 4: Time Scaling

If $x(t)$ is periodic with period T and fundamental frequency $\omega_0 = 2\pi/T$,

$$x(\alpha t)$$

where α is a positive real number, is periodic with period T/α and fundamental frequency $\alpha\omega_0$.

While the Fourier coefficients have not changed, the Fourier series representation has changed because of the **change in the fundamental frequency**.



Property 5: Multiplication

Suppose that $x(t)$ and $y(t)$ are both periodic with period T

$$x(t) \overset{\mathcal{FS}}{\leftrightarrow} a_k$$
$$y(t) \overset{\mathcal{FS}}{\leftrightarrow} b_k$$

Then

$$x(t)y(t) \overset{\mathcal{FS}}{\leftrightarrow} h_k = \underbrace{\sum_{l=-\infty}^{\infty} a_l b_{k-l}}_{\text{discrete-time convolution}}$$

Property 6: Conjugation and Conjugate Symmetry

If

$$x(t) \overset{\mathcal{FS}}{\leftrightarrow} a_k$$

Then

$$x^*(t) \overset{\mathcal{FS}}{\leftrightarrow} a^*_{-k}$$

For $x(t)$ real, i.e. $x(t) = x^*(t)$, the Fourier series coefficients will be **conjugate symmetric** $a_{-k} = a_k^*$.

Parseval's Relation

The total average power in a periodic signal equals the sum of the average powers in all of its harmonic components.

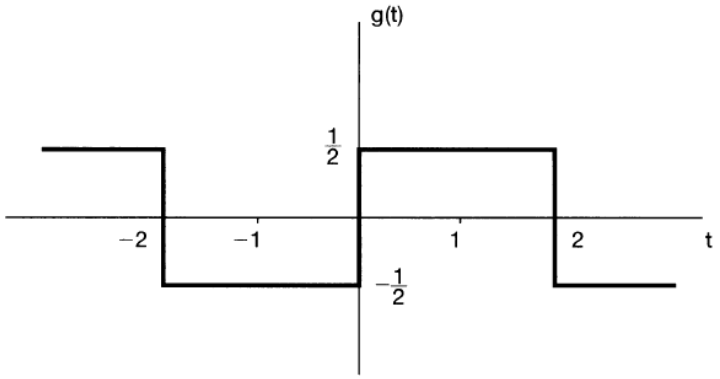
$$\frac{1}{T}\int_{T} |x(t)|^2 dt = \sum_{k=-\infty}^{\infty} |a_k|^2$$

average power (i.e., energy per unit time) in one period

Sum of average power in the *k*th harmonic component

Example 3.5

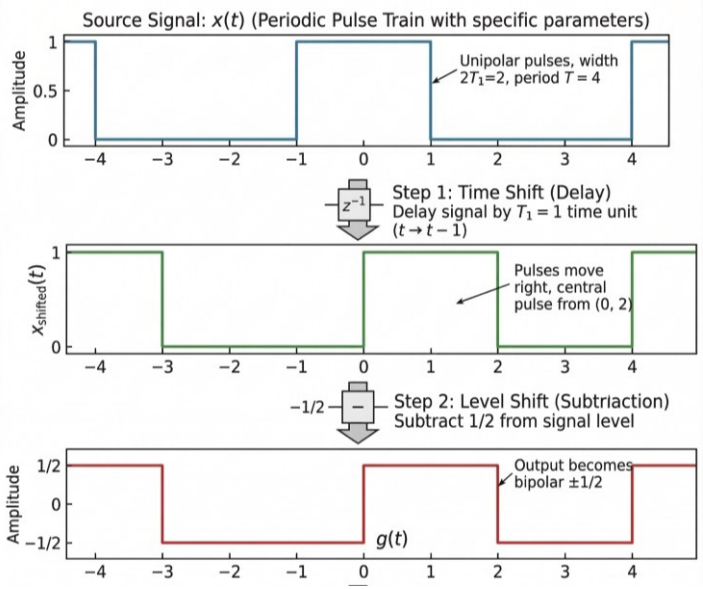
Consider the signal $g(t)$ with a fundamental period of 4, shown in the figure. Using suitable properties, determine the Fourier Series coefficients for $g(t)$.



<u>Solution:</u>

Refer to Example 3.4, we see that with $T = 4$, $T_1 = 1$,

$$g(t) = x(t-1) - \frac{1}{2}$$



Example 3.5 (Cont'd)

$$g(t) = x(t-1) - \frac{1}{2}$$

Using time-shift property, the Fourier coefficients of $x(t-1)$ may be expressed as

$$b_k = a_k e^{-jk(\pi/2)}$$

The Fourier coefficients of the dc offset in $g(t)$ ($-\frac{1}{2}$ term) are given by

$$c_k = \begin{cases} 0, & \text{for } k \neq 0 \\ -\frac{1}{2}, & \text{for } k = 0 \end{cases}$$

Applying the linearity property, the coefficients for $g(t)$ are

$$d_k = \begin{cases} a_k e^{-jk(\pi/2)}, & \text{for } k \neq 0 \\ a_0 - \frac{1}{2}, & \text{for } k = 0 \end{cases}$$

Example 3.5 (Cont'd)

From Example 3.4, $a_0 = \frac{2T_1}{T} = \frac{1}{2}$

$$a_k = \frac{\sin(k\omega_0 T_1)}{k\pi} = \frac{\sin(k\pi/2)}{k\pi}$$

$$d_k = \begin{cases} \frac{\sin(k\,\pi/2)}{k\pi} e^{-jk(\pi/2)}, & \text{for } k \neq 0 \\ 0, & \text{for } k = 0 \end{cases}$$

Fourier Series and LTI Systems

Input Signal:

$$x(t) = \sum_{k=-\infty}^{\infty} a_k e^{jk\omega_0 t}$$

Multiply each component by $H(jk\omega_0)$

Output Signal:

$$y(t) = \sum_{k=-\infty}^{\infty} a_k H(jk\omega_0) e^{jk\omega_0 t}$$

- Periodic signals decompose into a linear combination of harmonically related complex exponentials.
- The superposition property allows LTI systems to process each exponential component independently.
- The output is a new Fourier Series with modified coefficients.
- Input coefficients a_k multiply by the frequency response at that specific harmonic to create the output.

Example 3.6

Suppose that the periodic signal $x(t)$ discussed in Example 3.1 is the input signal to an LTI system with impulse response $h(t) = e^{-t}u(t)$, find $y(t)$.

Solution:

$$H(j\omega) = \int_{-\infty}^{\infty} h(\tau)e^{-j\omega\tau}d\tau = \int_{0}^{\infty} e^{-\tau}e^{-j\omega\tau}d\tau$$

$$= -\frac{1}{1+j\omega}e^{-(1+j\omega)\tau}\Big|_{0}^{\infty}$$

$$= \frac{1}{1+j\omega}$$

$\omega = k\omega_0$

Example 3.6 (Cont'd)

$$y(t) = \sum_{k=-3}^{3} b_k e^{jk2\pi t}$$

with $b_k = a_k H(jk2\pi)$ $\omega_0 = 2\pi$

$b_0 = 1,$

$b_1 = \frac{1}{4}\left(\frac{1}{1+j2\pi}\right),$ $b_{-1} = \frac{1}{4}\left(\frac{1}{1-j2\pi}\right)$

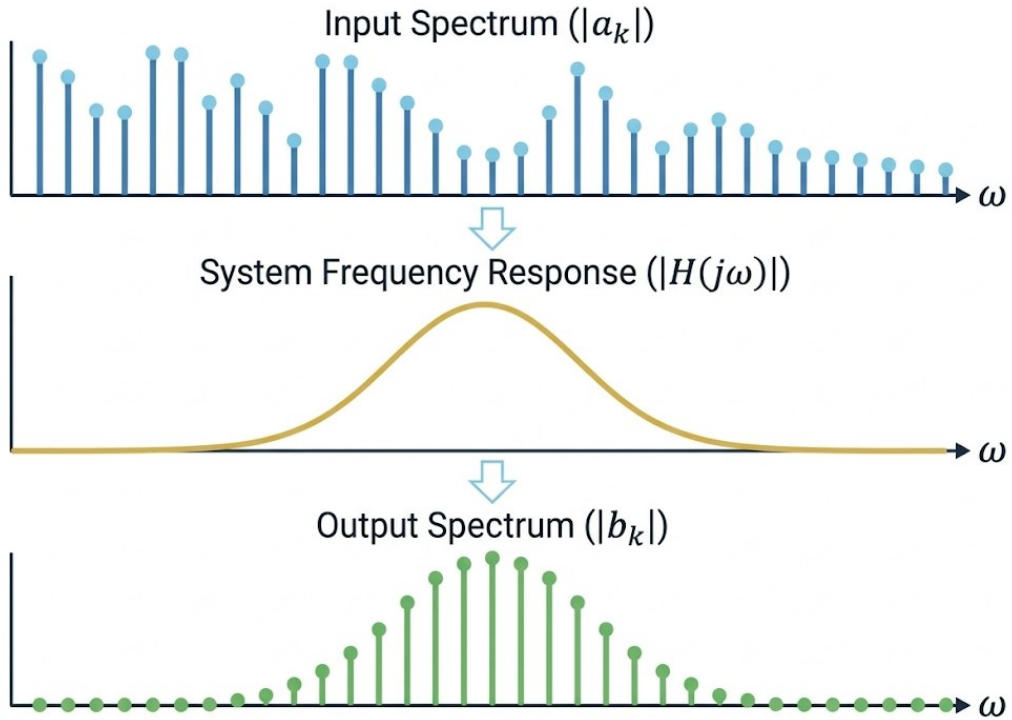
$b_2 = \frac{1}{2}\left(\frac{1}{1+j4\pi}\right),$ $b_{-2} = \frac{1}{2}\left(\frac{1}{1-j4\pi}\right)$

$b_3 = \frac{1}{3}\left(\frac{1}{1+j6\pi}\right),$ $b_{-3} = \frac{1}{3}\left(\frac{1}{1-j6\pi}\right)$

Filtering



- Filtering is the deliberate modification of the relative amplitudes or phases of a signal's frequency components.
- Used to **isolate desired information** or **suppress unwanted noise**.
- Implemented via LTI systems with mathematically engineered frequency responses.



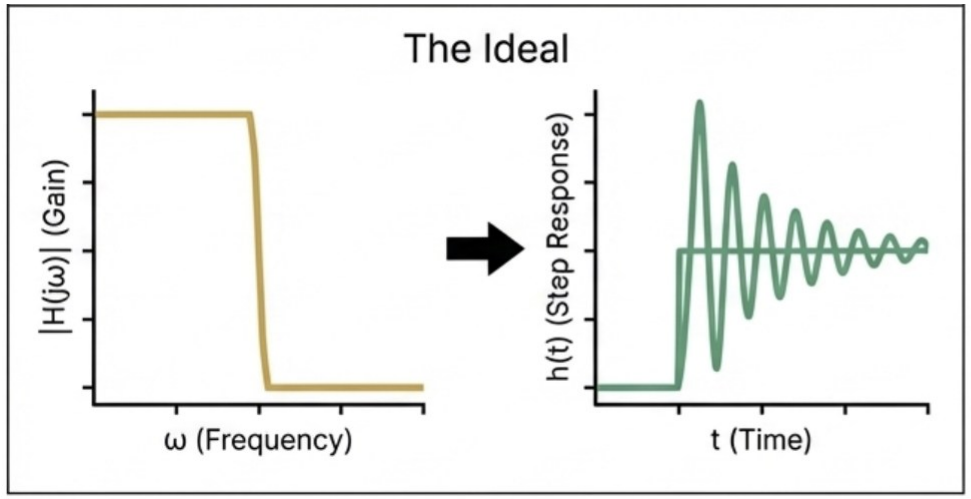
Two Filtering Paradigms

	Frequency-Shaping	Frequency-Selective
Goal	Modify the relative weighting of the spectrum.	Isolate specific frequency bands and eliminate others.
Function	Reshapes the signal curve (e.g., differentiating).	Acts as a gate (Passband vs. Stopband).
Example	Audio Equalizers, Edge Detectors	AM Radio Tuners, Noise Removal.

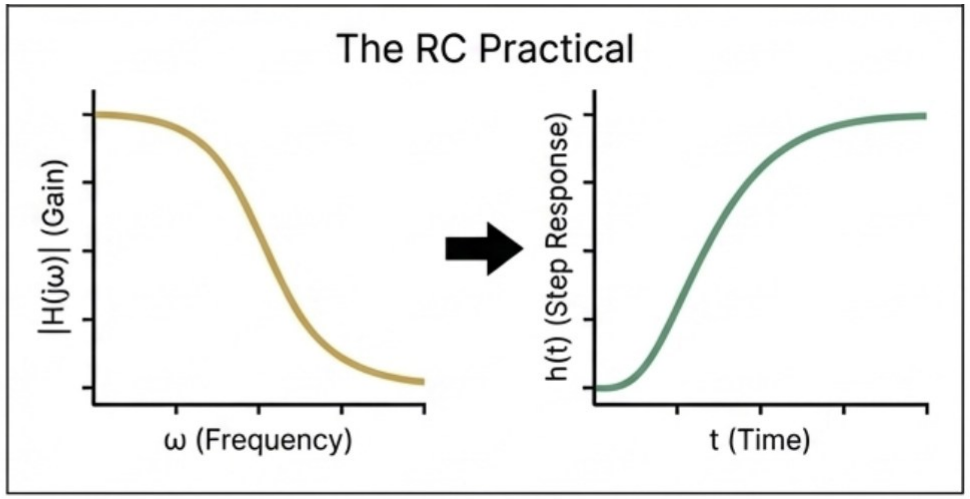
Both paradigms rely on the exact same underlying mathematics: $Y(j\omega) = H(j\omega)X(j\omega)$

The Time vs. Frequency Trade-Off

System design dictates an unavoidable tension between domains.



Aggressive frequency selectivity (sharp cutoffs) forces severe oscillatory ringing in the time-domain step response.



Smooth time-domain behavior (no overshoot) forces gradual, less precise frequency transitions.